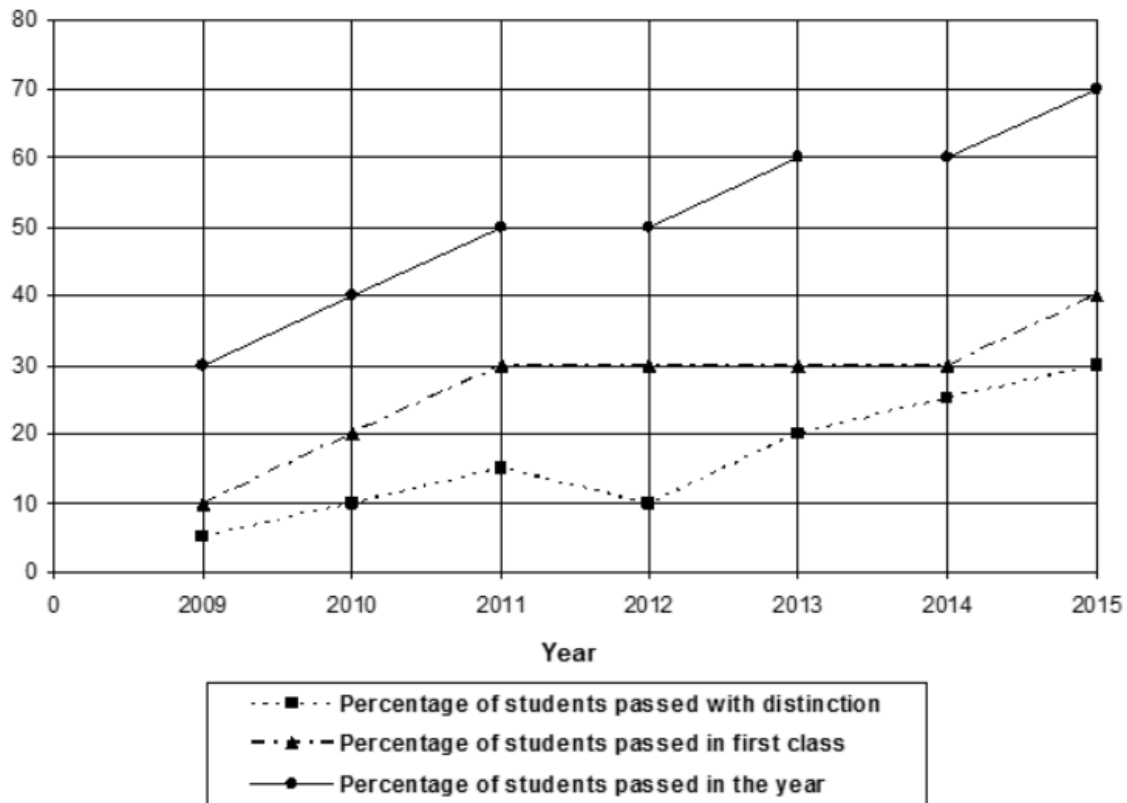


LINE GRAPH DI 9-9-26

Directions for questions 1 to 5: These questions are based on the line graph given below.



The above graph shows the distribution of results of students in a college during the period 2009-2015.

- (A) The total number of students appeared for the exam every year is the same i.e., 4000.
- (B) The students who passed with at least 80% marks are awarded a distinction.
- (C) The students who passed with 60% marks or more but less than 80% marks are awarded a first class.

Directions for questions 1 and 2: Select the correct alternative from the given choices.

The ratio of the number of students who failed in 2009 to those failed in 2012 is _____.

- ☐ a) $5/7$
- ☐ b) $2/7$
- ☐ c) $7/4$
- ☐ d) $7/5$

Solution:

The number of students who failed in 2009 = 70% of 4000 = 2800
And, that in 2012 = 50% of 4000 = 2000

$$\text{The ratio} = \frac{2800}{2000} = \frac{7}{5}$$

Choice (D)

The percentage increase in the number of students who passed in first class from 2009 to 2015 is _____.

- ☐ a) 300%
- ☐ b) 400%
- ☐ c) 225%
- ☐ d) 233.3%

Solution:

The number of students passing with first class in 2009 = 10% of 4000 = 400
And, that in 2015 = 40% of 4000 = 1600

$$\text{The percentage increase} = \frac{1200}{400} \times 100 = 300\%$$

Choice (A)

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

In which year is the number of students who passed in first class, the maximum?

Solution:

Observing the given graph, we can see that in 2015, the highest percentage of students passed in first class
Ans : (2015)

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

In 2012, what is the number of students who passed but got less than 60% marks?

Solution:

The number of students who passed in 2012
= 50% of 4000 = 2000
The number of students with 60% or more
= (10% + 30%) of 4000 = 1600
∴ The number of students who passed with less than 60% marks = 400.
Ans : (400)

The maximum number of students who passed the examination in any year is what percentage of the minimum number of students who passed in any year?

☐ a) $133\frac{1}{3}\%$

☒ b) $233\frac{1}{3}\%$ (Correct Answer - ✓)

☐ c) $42\frac{6}{7}\%$

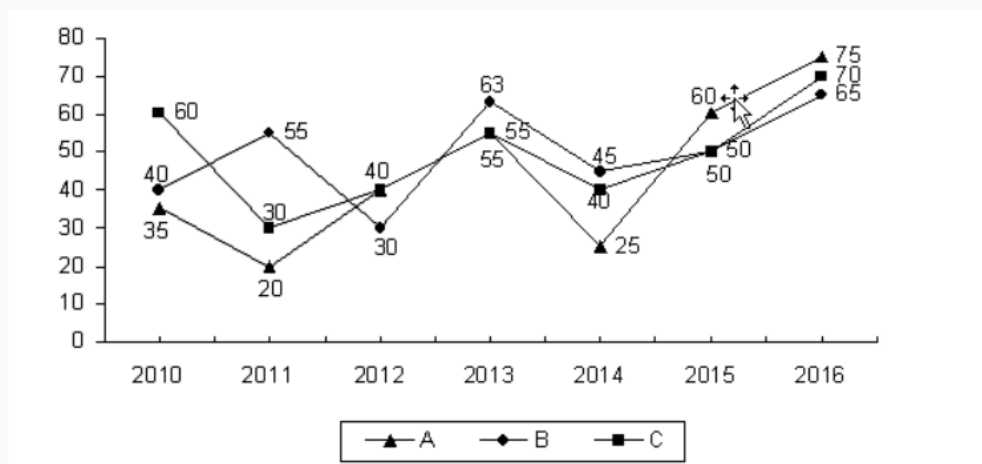
☐ d) $116\frac{2}{3}$

Solution:

$$\text{Required percentage} = \frac{70}{30} \times 100 = 233\frac{1}{3}\%$$

Choice (B)

Directions for questions 6 to 10: These questions are based on the following line graph.



Given above is the income of three chemical companies A, B, and C (in Rs. ₹ lakhs) for different years.

Income = Profit + Expenditure; Profit% =

$$\frac{\text{Profit}}{\text{Income}} \times 100$$

If the overall profit percentage of company A is 30%, then what is its overall profit (in Rs. ₹ lakh)?

☐ a) 90

☒ b) 93 (Correct Answer - ✓)

☐ c) 96

☐ d) 99

Solution:

Total income of A = 310.

Profit = 30% of 310.

$0.3 \times 310 = 93.$

Choice (B)

If the profit percentage of companies A and B in 2013 are 21% and 33% respectively, then the ratio of the profits of A and B in 2013 is _____.

☐ a) $\frac{4}{9}$

☐ b) $\frac{9}{5}$

☒ c) $\frac{5}{9}$ (Correct Answer - ✓)

☐ d) $\frac{7}{8}$

Solution:

Profit of A in 2013 = 0.21×55 .

Profit of B in 2013 = 0.33×63 .

$$\text{Ratio} = \frac{0.21 \times 55}{0.33 \times 63} = \frac{5}{9}.$$

Choice (C)

The expenditure of B in 2016 is equal to the income of B in 2012. What is its profit in 2016 (in Rs. ₹ lakh)?

☐ a) 20

☐ b) 30

☐ c) 32

☒ d) 35 (Correct Answer - ✓)

Solution:

$(\text{Expenditure of B})_{2016} = (\text{Income of B})_{2012}$

$65 - \text{Profit of B in 2016} = 30.$

$\therefore \text{Profit of B in 2016} = 35.$

Choice (D)

If the profit of company C increases by 10% in 2012 over its previous year, then, what is the ratio of expenditures of C in 2011 and 2012?

- ☐ a) $1\frac{1}{13}$
- ☐ b) $1\frac{3}{11}$
- ☐ c) $\frac{7}{9}$
- ☐ d) Cannot be determined

Solution:

Increase in profit is given. However, the values of the profit are unknown.
 $\therefore$ Neither the expenditure nor the ratio can be determined.

Choice (D)

In which year did any company record the highest percentage increase in its income over the previous year?

- ☐ a) 2014
- ☐ b) 2015
- ☐ c) 2016
- ☐ d) 2012

Show Correct Answer

Hide Solution

PREVIOUS

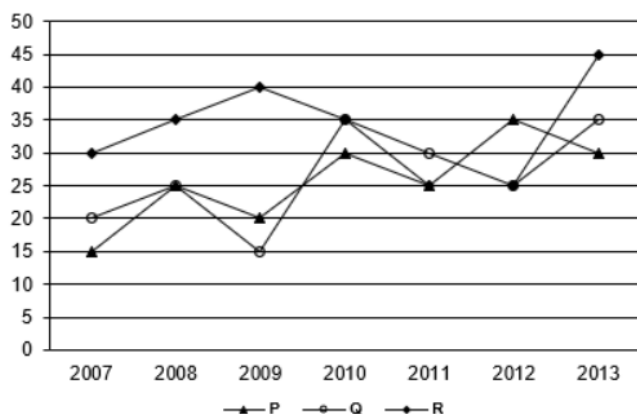
Solution:

Company A has the highest increase in its income from 2014 to 2015 i.e., $\frac{33}{25} \times 100$
 $= 140\%$.

Choice (B)

Directions for questions 11 to 15: Read the following line graph carefully to answer the questions that follow.

The line graph shows the income of three companies (in Rs. 000) A, B and C for the years 2007 to 2013.



$$\left(\frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \right) \times 100$$

If the expenditure of company P in 2013 is equal to 30% of the income of company R in 2010, what is the profit percentage (to the closest integer) of company P in 2013?

Your Answer: Unattempted

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

Income of company P in 2013 = 30

Income of company R in 2010 = 35

Expenditure of company P in 2013 = $\frac{30}{100} \times 35 = 10.5$

$\therefore$ Required percentage = $\frac{30 - 10.5}{10.5} \times 100$
= 186%

Ans : (186)

Directions for questions 11 and 12: Type in your answer in the input box provided below the question.

If the overall profit percentage of company A' is 20%, then find its overall profit (in Rs.).

Solution:

Overall income of company
 $P = 15 + 25 + 20 + 30 + 25 + 35 + 30 = 180$

$$\text{Profit \%} = \frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \times 100$$

$$\Rightarrow 20 = \frac{180 - \text{Expenditure}}{\text{Expenditure}} \times 100$$

$$\Rightarrow \text{Expenditure} = 150$$

$$\therefore \text{Required profit} = (180 - 150) \times 1000 \\ = ₹30,000.$$

Ans : (30,000)

If the profit percentage of company Q in 2010 is 40% and company C in 2009 is 60%, then find the ratio of their profits.

- ☐ a) 2 : 3
- ☐ b) 3 : 2
- ☐ c) 3 : 4
- ☐ d) 5 : 4

Solution:

Given,
Profit percentage and income of company
Q in 2010 is 40% and 35,000 respectively.

$$40 = \frac{35 - \text{Expenditure}}{\text{Expenditure}} \times 100 \Rightarrow \text{Expenditure} = 25$$

$$\therefore \text{Profit of company Q in 2010} \\ = 35 - 25 = 10,000$$

Given, the profit percentage and income of company R in 2009 was 60% and 40,000 respectively.

$$60 = \frac{40 - \text{Expenditure}}{\text{Expenditure}} \times 100$$

$$\Rightarrow \text{Expenditure} = 25,000$$

$$\therefore \text{Profit Company R in 2009} \\ = 40 - 25 = 15,000$$

$$\therefore \text{Required ratio} = 10 : 15 = 2 : 3.$$

Choice (A)

What is the approximate average income of company C in the given years

- ☐ a) 28
- ☐ b) 26
- ☐ c) 32
- ☐ d) 34

Solution:

$$\begin{aligned} \text{Required average} \\ = \frac{30+35+40+35+25+25+45}{7} = 33.57 = 34. \end{aligned}$$

Choice (D)

Directions for questions 13 to 15: Select the correct alternative from the given choices.

If the income of company A in 2008 is equal to the expenditure of company B in 2010, what is the profit percentage of company B in 2010?

- ☐ a) 45%
- ☐ b) 60%
- ☐ c) 40%
- ☐ d) 35%

Solution:

$$\begin{aligned} \text{Income of company P in 2008} &= 25 \\ \Rightarrow \text{Expenditure of company Q in 2010} &= 25 \\ \therefore \text{Required percentage} &= \frac{35-25}{25} \times 100 \\ &= 40\%. \end{aligned}$$

Choice (C)